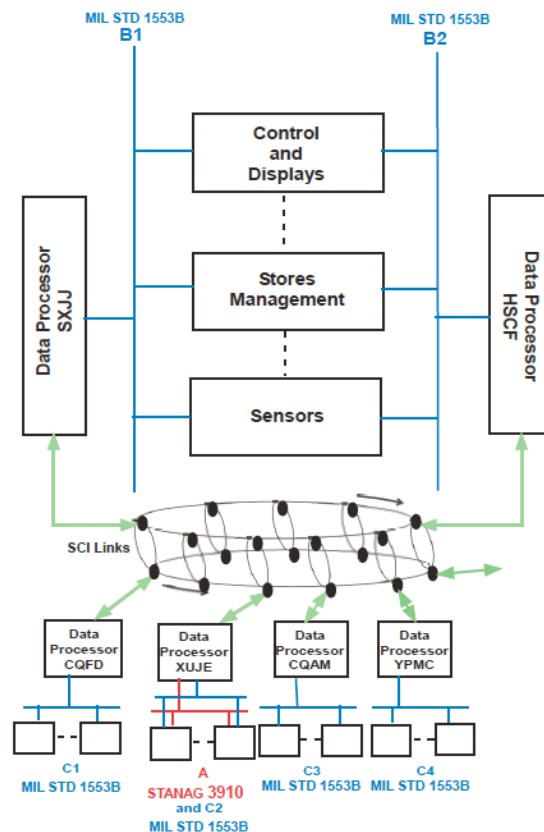


MIL STD 1553B Application

We consider the following communication architecture, which is a representative architecture of the Rafale aircraft. This architecture is based on 6 MIL STD 1553B buses connecting the calculators, sensors and actuators and a SCI core network (a high-speed mesh network (200Mbps per link)).



Objective: Performance analysis of the 1553B network (the most heavily loaded network) in terms of end-to-end communication delays.

The end-to-end communication time of a message consists of:

- d1: The crossing time of the layers
- d2: The access delay that depends on the MAC protocol
- d3: The transmission delay
- d4: propagation delay
- d5: The reception delay

P.S : We assume in our case that the delays d1, d4 and d5 are negligible

On the MIL STD 1553B B1 network, the master is SXJJ. The characteristics of the messages transmitted on the network (name, frequency, size (words), transmitter, receiver) are described in xmlB1-periodic.xml (to download under LMS).

In order to achieve our objective, we will follow the following steps through implementing the different functions required for each step. The choice of the programming language is free.

Step1: Interpretation of input data

- Download the .xml file under LMS, containing the characteristics of the messages
- Parse the xml file to have the list of messages and their characteristics.

Step2: Computation of the transmission delay (d3)

- Based on the sens of communication, compute the total size of each message.
- Deduce the transmission delay of each message by taking into account a link speed of 1Mbps

Step3: Performance analysis in terms of end-to-end delays and access delays (d2)

- Calculate the end-to-end delay of each message.
- Deduce the access times to the medium (d2)
- Conclude on the schedulability of each message

Step 4: Output xml file generation

Generate an output xml file that contains for each message, in addition to these initial characteristics, the following information:

- Transmission delay (<DT>)
- Access delay to the medium (<DMAC>)
- end-to-end delay (<DBEB>)
- schedulability test (<Test>)